

Sea Level Rise and Flood Risk in Long Island, New York

Li, J.¹ & Marsellos, A.E.²

Roslyn High School, 475 Round Hill Rd, Roslyn Heights

²Dept. Geology, Environment and Sustainability, Hofstra University, Hempstead, NY, USA



Abstract:

Long Island, New York is experiencing increasing sea level rise resulting from climate change. Using a GIS platform (Global Mapper v.24.1) we simulated sea level rise scenarios at 0.5 m, 1 m, 2 m, and 4 m to evaluate future flood risk along Long Island's shoreline. A Digital Elevation Model (DEM) and shapefiles of Long Island implemented to analyze the potential future coastal areas. Visualizing the scenarios of potential flooding indicated significant inundation risk for low-lying areas, especially those near Fire Island and South Shore coastal regions along the southern shoreline of Long Island. By analyzing all flooding profiles with coastal infrastructure in place, we were able to evaluate communities who are at risk for flooding over the next 50–200 years. Our analysis provides insights that even minor variations of sea level rise (0.5-1 m)) can cause significant shoreline retreat and land loss. Due to increasing global warming and climate change continuing to accelerate impacts and our ability to assess impacts of sea level rise at a regional scale, it is especially crucial to plan and prepare strategies for adapting and protecting coastal communities, regional populations, and providing reduced environmental and economic damages.

Background Information:

- Sea level in New York already increased by approximately ~9 inches from 1900
- Projections: +2 feet by 2070, +3–5 feet by 2100 (depending on emissions)
- Low-lying barrier islands on Long Island's South Shore (Fire Island, Jones Beach) are especially vulnerable
- Regular storms and hurricanes contribute to coastal flooding
- Use of Global Mapper + DEM data allows us to visualize the impacts that the water rise has on coastal communities



Figure 1: Base map of Long Island, New York. Used as a geographic reference for sea level rise simulations

Method/Materials:

- **Global Mapper (GIS software)**
 - Simulating sea level rise scenarios
- **Digital Elevation Model (SRTM data)**
 - Used to model terrain
- **Shapefiles (Long Island boundary & shoreline)**
 - Overlaid for geographic accuracy
- **Sea Level Rise Projections**
 - 0.5 m (50 yrs), 1.0 m (100 yrs), 2.0 m (150 yrs), 4.0 m (200 yrs)
- **Maps created in both 2D and 3D for visualization**

Results:

- **At 0.5 m rise:** noticeable flooding along Fire Island and low-lying South Shore
- **At 1.0 m rise:** barrier islands show significant inundation, with inland encroachment
- **At 2.0 m rise:** severe land loss across much of the South Shore
- **At 4.0 m rise:** large portions of Long Island's southern coastline underwater
- **Maps demonstrate clear spatial differences in vulnerability**

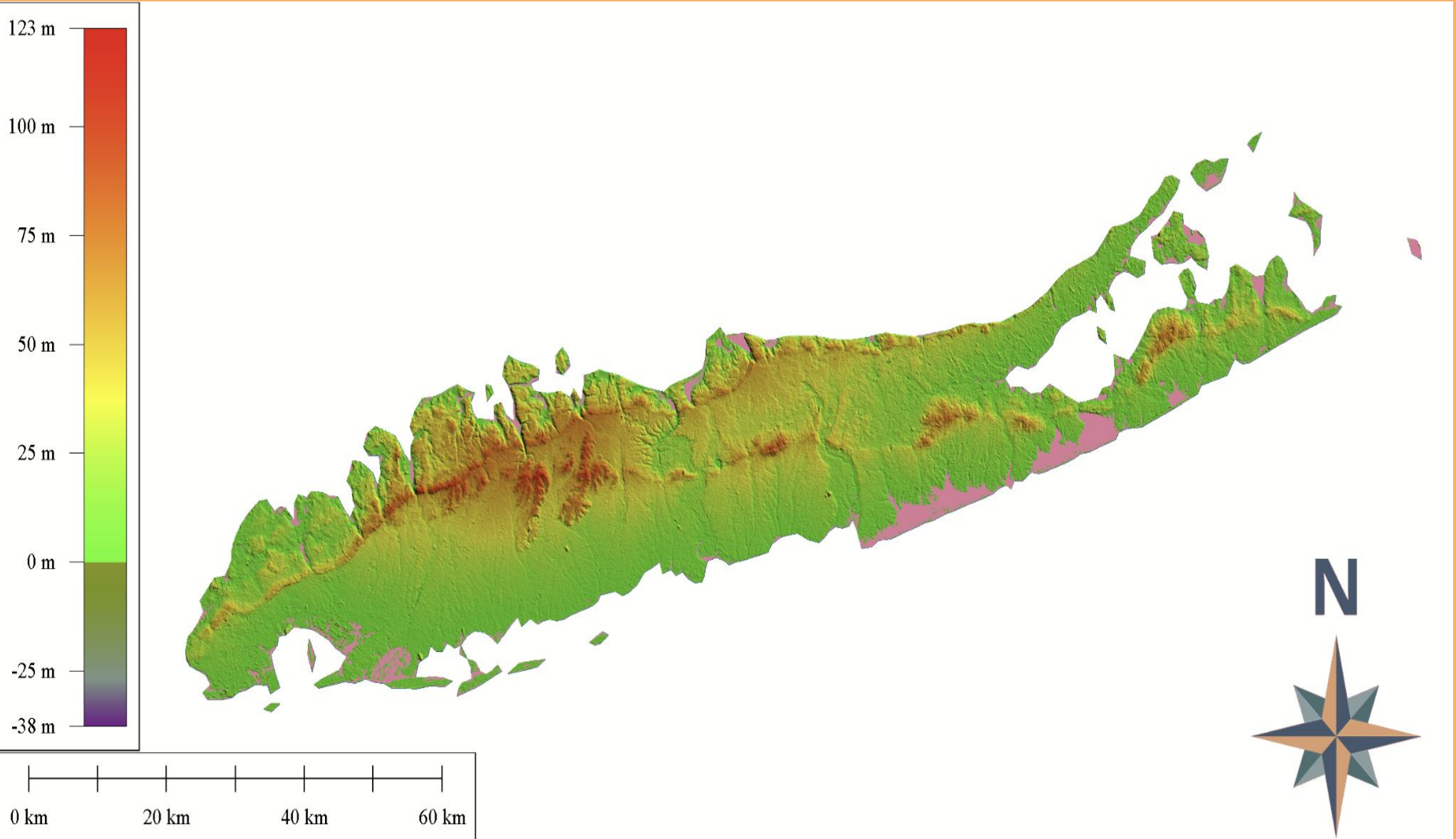


Figure 2: Simulated flooding at 0.5 m sea level rise (approx. 50 years). Low-lying South Shore areas, including parts of Fire Island, show initial flooding.

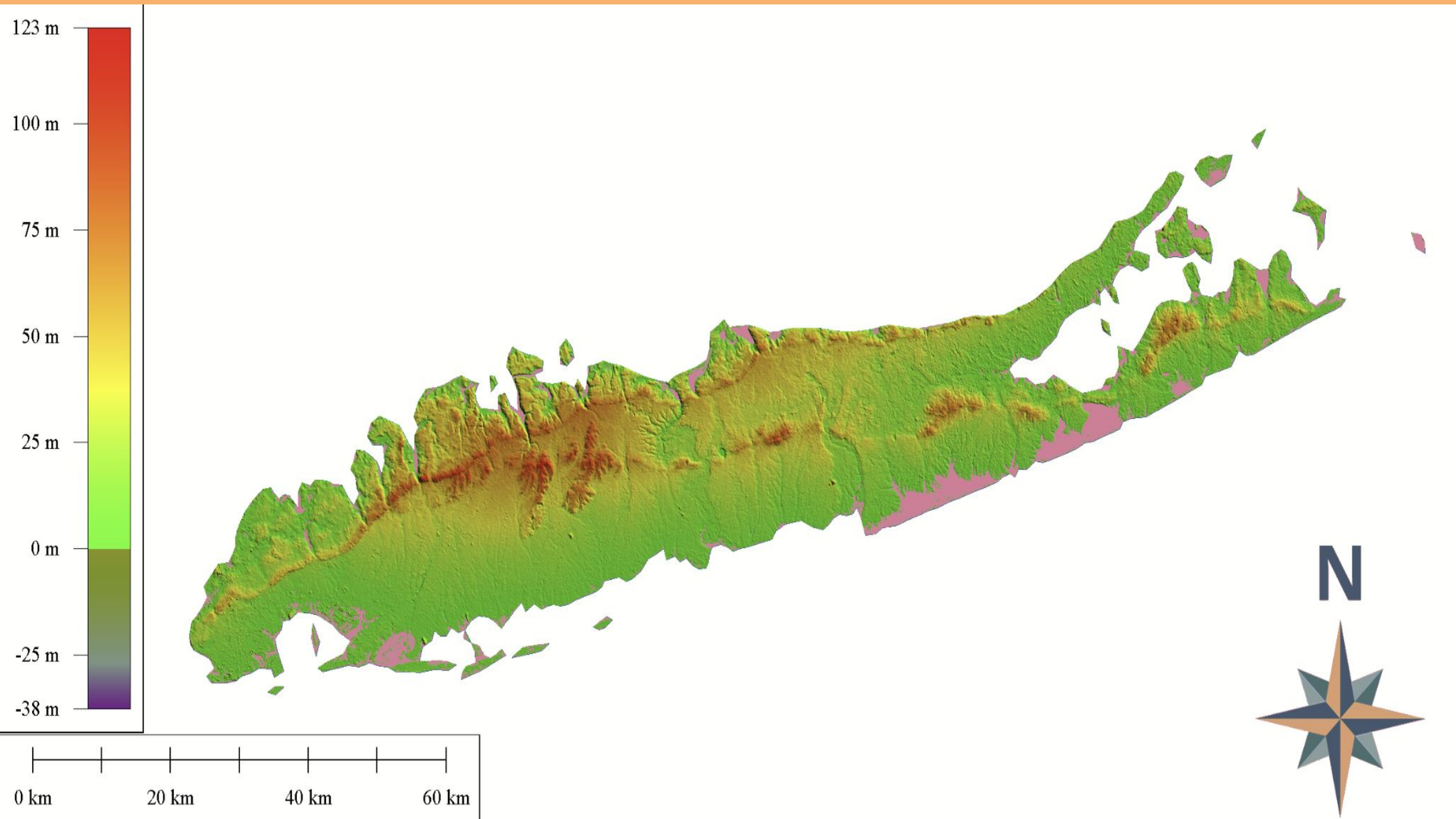


Figure 3: Simulated flooding at 1.0 m sea level rise (approx. 100 years). Significant flooding of barrier islands and encroachment into inland communities.

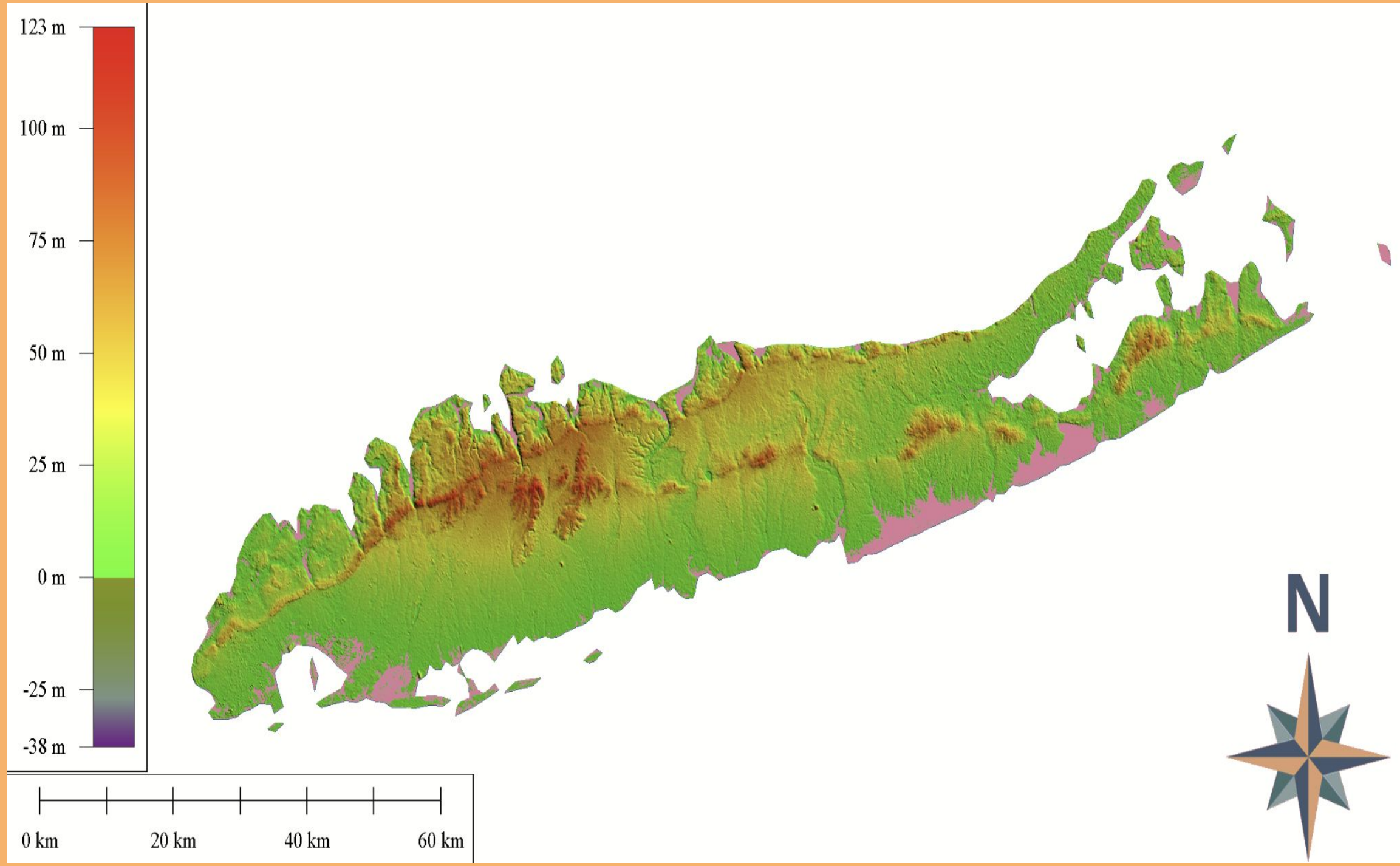


Figure 4: Simulated flooding at 2.0 m sea level rise (approx. 150 years). Widespread flooding across the South Shore, with land loss visible.

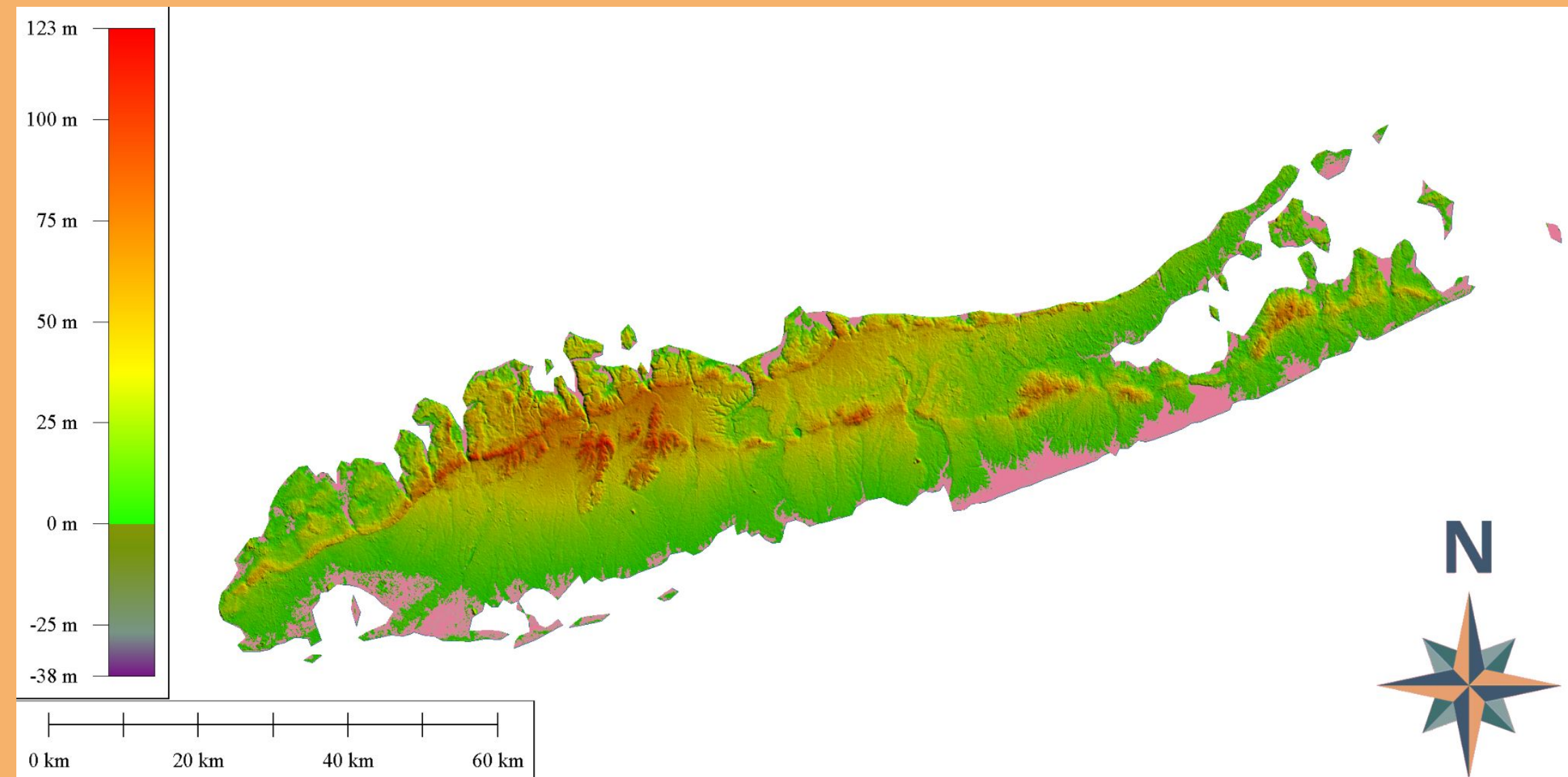


Figure 5: Simulated flooding at 4.0 m sea level rise (approx. 200 years). Large portions of Long Island's southern coastline are underwater, indicating long-term risk.

Discussion/Conclusion:

- Even relatively small rises in sea level presents serious risks for coastal Long Island
- Fire Island and barrier beaches act as the first line of defense, but when they are eroded, an increased risk of flooding is occurring farther inland
- Infrastructure (roads, homes, utilities) at risk in South Shore communities
- Adaptation strategies (sea walls, managed retreat, wetland restoration) will be essential
- GIS-based modeling is an effective way to visualize and communicate climate risks

Acknowledgements:

We would like to thank Dr. Antonios E. Marsellos and the Geology, Environment and Sustainability department at Hofstra University for providing us with the necessary tools to conduct this research.

Credit Authorship Contribution Statement:

Li, J: GIS simulations in Global Mapper, shapefile and DEM integration, map creation (Figures 1–5), writing – Abstract, Background, Methods, Results, Discussion; Marsellos, A.E: supervision/tutoring, technical guidance with GIS and sea level rise modeling, editing.

References:

USGS. Long Island Topography. U.S. Geological Survey. March 27, 2024. Long Island Topography | U.S. Geological Survey ([usgs.gov](https://www.usgs.gov))
New York State Climate Impacts Assessment. (2024). *Precipitation and Sea Level Rise Projections for New York*. March 27, 2024. <https://nysclimateimpacts.org>
NOAA. *Sea Level Rise Viewer*. National Oceanic and Atmospheric Administration. March 27, 2024. <https://coast.noaa.gov/slr>
Blue Marble Geographics. Global Mapper v.24.1. GIS software used for sea level rise simulations. <https://www.blumarblegeo.com/global-mapper/>